CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 23: Trigonometrical Ratios of Standard Angles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

2. Evaluate $\tan^2 60^\circ + 4\cos^2 45^\circ + 3\sec^2 30^\circ$.

3. Evaluate $\dfrac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$.

4. Evaluate $\cos^2 30^\circ - \sin^2 30^\circ$ and name the standard ratio it equals.

5. If $A = 30^\circ$, verify that $\cos 2A = 1 - 2\sin^2 A$.

6. Find the acute angle θ if $\sqrt{3}\tan\theta = 1$.

7. Find θ if $\sin 2\theta = \dfrac{\sqrt{3}}{2}$ and 2θ is acute.

8. Find A and B if $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \dfrac{1}{\sqrt{3}}$, with $A+B$ acute and $A > B$.

9. Solve $4\cos^2\theta = 3$ for an acute angle.

10. In triangle PQR, $\angle Q = 90^\circ$, $\angle P = 60^\circ$ and $PQ = 5$ cm. Find QR and PR.

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Answer Key

1. 1
2. $3 + 2 + 4 = 9$
3. $\dfrac{\frac{3}{2} - \frac{2}{\sqrt{3}}}{\frac{3}{2} + \frac{2}{\sqrt{3}}} = \dfrac{43 - 24\sqrt{3}}{11}$
4. $\dfrac{1}{2} = \cos 60^\circ$
5. Both sides equal $\dfrac{1}{2}$
6. 30°
7. 30°
8. $A = 45^\circ, B = 15^\circ$
9. 30°
10. $5\sqrt{3}$ cm and 10 cm

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